T TRIAGE

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TRIAGE Bottleneck Hunt

Small listed companies inside supply chokepoints: asymmetric setups, base rates and kill conditions

Setups	Base rate	Rules
15 WATCH calls, hash-chained	7.0% of sub-$300m entries touched 10x in 12m	public data · research only · not advice

As of 23 Oct 2026 (SGT). Research only, not investment advice, paper ledger only. These are asymmetric setups, not promises. We do not claim 100x for anything.

01 The honest base rate first

Universe: 45 chokepoint-adjacent small and micro caps (antimony, tungsten, rare earths, graphite, scandium, helium, InP/optics, isotopes, probe cards, lithium) with prices back to Jan 2023. Each month from Jan 2023 to Oct 2025 is an entry; we look 12 months ahead.

BUCKET	ENTRIES	NAMES	TOUCHED 2X	TOUCHED 5X	TOUCHED 10X	STILL ≥10X AT 12M	ENDED ≥2X	LOST ≥50%	MEDIAN 12M RETURN	MEDIAN MAX DRAWDOWN
Entry cap < $300m (approx.)	985	36	44.1%	16.4%	7.0%	3.2%	27.9%	13.8%	8.7%	-54.4%
All entries	1530	45	46.1%	14.2%	4.9%	2.3%	27.1%	15.2%	12.1%	-55.0%

Best 12-month peak multiple per name (sub-$300m entries): TLO.TO 108.9x, AXTI 89.7x, KULR 32.0x, UAMY 28.6x, UCU.V 20.9x, ASPI 16.3x, EQR.AX 16.1x, AREC 15.7x, LRV.AX 14.9x, AAOI 12.7x, TUN.L 11.9x, ARA.TO 10.5x.

Read this before anything else. The universe was chosen in 2026 from names we remember, so it is biased towards names that moved (survivorship and recall bias). Delisted and failed names are under-represented. Entry market caps are approximated using today's share count, which understates the true small-cap count for heavy diluters. The true base rate is lower than this table. A 100x in months is not in the distribution we can measure.

02 Method

Top-down chokepoints (MOFCOM export controls, DoD DPA Title III/DLA awards, NWF/EXIM finance, NDAA sourcing rules) → small listed firms that hold the scarce asset, permit or qualification → evidence graded A (government document), B (company primary filing) or C (company claim or trade press, not re-verified) → score.

Score (0–5) = 25% chokepoint share + 20% revenue sensitivity + 15% size + 15% evidence grade + 10% liquidity + 15% not-crowded. The weights are judgement. Contrarian check: 12-month return >200%, top 20% of 52-week range, 3-month return >60%. Quant Validation: every name is VETOED as recommendation-grade because no backtested KEPT signal underlies it, so all are paper WATCH calls.

The dated catalyst for the whole metals cluster: MOFCOM Announcement 72/2025 suspends the US-specific ban on gallium, germanium, antimony and superhard materials (Ann. 46/2024 Art. 2) **only until 27 Nov 2026**. Art. 1 (no dual-use items to US military end users) stays in force. Source: mofcom.gov.cn/zcfb/blgg/gg/202...4f36dc5.html

03 Top 15

#	COMPANY	CHOKEPOINT	MCAP $M	ADV $K	12M	SCORE	GRADE	CROWDING	CATALYST
1	United States Antimony (UAMY)	Antimony (defence: primers, munitions; flame retardants)	465.5	19624	-76%	4.25	A	Not crowded on price	2026-11-27: MOFCOM suspension of the US-specific antimony/gallium/germanium ban expires unless renewed
2	Larvotto Resources (LRV.AX)	Antimony concentrate (non-China mine supply)	290.4	1067	-25%	4.2	B	Not crowded on price	2026-11-27: MOFCOM suspension expiry; first quarterly production report (Jan 2027)
3	NioCorp Developments (NB)	Scandium (Al-Sc alloys; China-controlled since Apr 2025) + niobium	431.0	8914	-65%	4.2	A	Not crowded on price	2027-06-30: EXIM final commitment decision (judgement: H1 2027)
4	Ucore Rare Metals (UCU.V)	Rare-earth separation (Dy/Tb/Sm outside China)	210.6	506	-70%	4.0	B	Not crowded on price	2026-12-31: H2-2026 Early Production readiness deadline in Louisiana
5	Guardian Metal Resources (GMET.L)	Tungsten (US domestic mine supply)	211.5	274	-26%	3.9	A	Not crowded on price	2026-11-27: MOFCOM step; Pilot Mountain permitting milestones 2027
6	Graphite One (GPH.V)	Natural graphite + anode material (China export controls since Dec 2023)	111.9	29	-35%	3.85	A	Not crowded on price	2026-11-27: MOFCOM graphite end-use review suspension expiry; EXIM application 2026
7	LightPath Technologies (LPTH)	Germanium-free infrared optics (BlackDiamond chalcogenide glass)	485.7	20838	-6%	3.85	B	Not crowded on price	2027-12-31: NDAA germanium sourcing restriction takes effect (Dec 2027)
8	Aclara Resources (ARA.TO)	Heavy rare earths (Dy/Tb) from ionic clays, US separation	460.4	163	-3%	3.8	A	Not crowded on price	2026-12-31: Q4-2026 Dy/Tb pilot specification results
9	Pulsar Helium (PLSR.V)	Helium (US federal reserve sold; Qatar/Russia concentration)	190.5	156	40%	3.75	B	Not crowded on price	2027-06-30: Up to six more wells Q1–Q2 2027
10	Japan Electronic Materials (6855.T)	Probe cards for DRAM/HBM wafer test	580.0	26038	69%	3.6	B	Not crowded on price	2026-11-10: 1H FY3/27 results (judgement: early Nov, check TDnet calendar)
11	American Resources (ReElement) (AREC)	Rare-earth refining (chromatographic) in the US	118.7	2201	-70%	3.55	C	Not crowded on price	2027-03-31: ReElement financing (judgement)
12	ASP Isotopes (ASPI)	Isotope enrichment (Mo-100, Yb-176; HALEU ambition)	316.6	8458	-77%	3.5	C	Not crowded on price	2027-06-30: Commercial isotope deliveries (judgement)
13	Tungsten West (TUN.L)	Tungsten (APT for tooling, armour, munitions)	767.2	1903	235%	3.48	A	Partly crowded	2027-03-31: Hemerdon full commissioning target (Q1 2027)
14	Tri Chemical Laboratories (4369.T)	Hafnium high-k ALD precursors for DRAM	617.9	9934	0%	3.45	C	Not crowded on price	2026-12-12: Q3 FY1/27 results (judgement: mid-Dec)
15	Ionic Rare Earths (IXR.AX)	Magnet recycling + Uganda ionic clay (Makuutu)	101.2	295	17%	3.41	C	Not crowded on price	2027-03-31: Belfast scale-up (judgement)

1. UNITED STATES ANTIMONY (UAMY)

- **Chain:** US DoD National Defense Stockpile → antimony metal ingots → USAC, operator of North America's only two antimony smelters
- **Verified fact:** DLA sole-source 5-yr IDIQ SP8000-25-D-0007, max $245m (no obligation beyond $5k minimum); first delivery order ~$10m (315,000 lb). China's US antimony ban (MOFCOM Ann. 46/2024 Art. 2) is suspended only until 27

Nov 2026 (Ann. 72/2025).

- **Catalyst:** 2026-11-27: MOFCOM suspension of the US-specific antimony/gallium/germanium ban expires unless renewed
- **Key risks:** IDIQ max is not revenue; feed-ore sourcing (Mexico/Bolivia); China renews the suspension → price normalises. Dilution risk: MEDIUM.
- **Kill condition:** MOFCOM renews the suspension AND antimony price (Rotterdam) falls >30% from 23 Oct 2026; or no second DLA delivery order by 30 Jun 2027
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** sec.gov/Archives/edgar/data/10...0xex10d1.htm · sec.gov/Archives/edgar/data/10.../uamy_8k.htm · mofcom.gov.cn/zcfb/blgg/gg/202...4f36dc5.html

2. LARVOTTO RESOURCES (LRV.AX)

- **Chain:** Western antimony smelters/trader Wogen → Sb concentrate → Hillgrove (NSW), production started Aug 2026
- **Verified fact:** Production commenced Aug 2026; first concentrate shipped Oct 2026; Wogen exclusive offtake for 7 years with US$4m prepayment.
- **Catalyst:** 2026-11-27: MOFCOM suspension expiry; first quarterly production report (Jan 2027)
- **Key risks:** ramp-up recoveries; single offtaker sets terms; antimony price mean-reversion. Dilution risk: MEDIUM.
- **Kill condition:** Quarterly report shows concentrate output <50% of nameplate two quarters running
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** larvottoresources.com/wp-conte...61341464.pdf · larvottoresources.com/wp-conte...61349174.pdf · larvottoresources.com/wp-conte...61241120.pdf

3. NIOCORP DEVELOPMENTS (NB)

- **Chain:** Defence prime (Al-Sc aerospace alloys) → scandium oxide → Elk Creek, Nebraska
- **Verified fact:** DoD DPA Title III up to $10m (Aug 2025); pursuing up to $800m EXIM debt (not committed).
- **Catalyst:** 2027-06-30: EXIM final commitment decision (judgement: H1 2027)
- **Key risks:** ~$1bn+ capex; financing not committed; history of delays and dilution. Dilution risk: HIGH.
- **Kill condition:** EXIM declines or no final commitment by end-2027
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** war.gov/News/Releases/Release/...o-master-al/ · niocorp.com/u-s-department-of-...ources-corp/

4. UCORE RARE METALS (UCU.V)

- **Chain:** US DoD magnet supply chain → separated REE oxides → RapidSX plant, Alexandria, Louisiana
- **Verified fact:** DoD OT Agreement total US$22.4m (non-repayable); target 'Early Production' readiness in Louisiana in H2 2026; DPAS DO-B8 rating.
- **Catalyst:** 2026-12-31: H2-2026 Early Production readiness deadline in Louisiana
- **Key risks:** RapidSX unproven at commercial scale; feedstock sourcing; deadline slip. Dilution risk: HIGH.
- **Kill condition:** No early-production announcement by 31 Mar 2027
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** ucore.com/ucore-executes-us18-...-processing/ · ucore.com/FinancialStatements/..._Q2_2025.pdf

5. GUARDIAN METAL RESOURCES (GMET.L)

- **Chain:** US DoD (DPA Title III) → WO3 → Pilot Mountain, Nevada
- **Verified fact:** DoD DPA Title III $6.2m award (Jul 2025); PFS June 2026: 8-year mine, 15,916 t WO3, first production targeted late 2028 subject to permitting.
- **Catalyst:** 2026-11-27: MOFCOM step; Pilot Mountain permitting milestones 2027

- **Key risks:** pre-production, 2028 at earliest; financing dilution; permitting. Dilution risk: HIGH.
- **Kill condition:** Permitting timeline slips beyond 2029 or equity raise below 50% of 23 Oct 2026 price
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** war.gov/News/Releases/Release/...on-of-tungs/ · sec.gov/Archives/edgar/data/20...ultsfina.htm

6. GRAPHITE ONE (GPH.V)

- **Chain:** US battery and defence users → anode active material → Graphite Creek (Alaska) + Ohio anode plant
- **Verified fact:** DoD DPA Title III $37.5m (Jul 2023); EXIM non-binding LoIs total $2.07bn; formal application expected 2026.
- **Catalyst:** 2026-11-27: MOFCOM graphite end-use review suspension expiry; EXIM application 2026
- **Key risks:** LoIs non-binding; Arctic mine capex; EV demand. Dilution risk: HIGH.
- **Kill condition:** No EXIM application filed by 30 Jun 2027
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** defense.gov/News/Releases/Rele...ning-and-pr/ · graphiteoneinc.com/exim-bank-i...-07-billion/

7. LIGHTPATH TECHNOLOGIES (LPTH)

- **Chain:** US DoW EO/IR & counter-UAS systems → IR lenses → BlackDiamond (licensed from the US Naval Research Lab)
- **Verified fact:** Funded DoW CLEAR prototype agreement (Sep 2026); 2026 NDAA adds germanium to covered-material restrictions with a Dec 2027 sourcing deadline; >$19m orders from one C-UAS customer since 1 Jul 2026.
- **Catalyst:** 2027-12-31: NDAA germanium sourcing restriction takes effect (Dec 2027)
- **Key risks:** customer concentration; already re-rated; CLEAR value undisclosed. Dilution risk: MEDIUM.
- **Kill condition:** No production follow-on from CLEAR by end-2027
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** finance.yahoo.com/technology/a...3100543.html · newswire.com/news/lightpath-te...w-on-optical

8. ACLARA RESOURCES (ARA.TO)

- **Chain:** Magnet makers → Dy/Tb oxide → Project Dynamo, Vinton LA ($494m) fed by Chile/Brazil clays
- **Verified fact:** Basic engineering complete; ~200 t/yr Dy oxide and 35 t/yr Tb oxide planned; Dy/Tb spec results from the Virginia Tech pilot expected Q4 2026; construction to H2 2028.
- **Catalyst:** 2026-12-31: Q4-2026 Dy/Tb pilot specification results
- **Key risks:** $494m capex vs ~C$645m cap → large financing; Chile permitting history. Dilution risk: HIGH.
- **Kill condition:** Pilot misses Dy/Tb spec or the Q1-2027 site prep slips
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** opportunitylouisiana.gov/news/...in-louisiana · enr.com/articles/63926-494m-ra...construction

9. PULSAR HELIUM (PLSR.V)

- **Chain:** Semiconductor fabs / MRI / space → liquid helium → Topaz, Minnesota (high-grade primary helium)
- **Verified fact:** Accepted Chart Industries $85.5m firm proposal for a helium liquefaction plant (Oct 2026); first production targeted Q4 2027; no helium sales agreement disclosed.
- **Catalyst:** 2027-06-30: Up to six more wells Q1–Q2 2027
- **Key risks:** no offtake yet; $85.5m plant vs ~C$267m cap → dilution; helium price has softened. Dilution risk: HIGH.
- **Kill condition:** Appraisal wells fail to confirm flow rates
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** pulsarhelium.com/wp-content/up...hart-RNS.pdf · pulsarhelium.com/projects/topaz/

10. JAPAN ELECTRONIC MATERIALS (6855.T)

- **Chain:** Micron (HBM/DRAM) → memory probe cards → JEM (Micron entities ~31% of FY3/26 sales)
- **Verified fact:** Annual securities report FY3/26: Micron Memory Japan ¥4.567bn (15.6%) and Micron Memory Taiwan ¥4.495bn (15.3%) of sales.
- **Catalyst:** 2026-11-10: 1H FY3/27 results (judgement: early Nov, check TDnet calendar)
- **Key risks:** FormFactor/MJC competition; memory cycle; Micron dependence. Dilution risk: MEDIUM.
- **Kill condition:** Micron share of sales falls below 20% or 1H order book declines y/y
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** jem-net.co.jp/images/top/pdf/2...okokusho.pdf · edinetdb.jp/company/E02043/main-customers

11. AMERICAN RESOURCES (REELEMENT) (AREC)

- **Chain:** Magnet makers → separated oxides → ReElement Technologies (Indiana)
- **Verified fact:** Company claims (not re-verified this pass): ReElement refining capacity and offtake MOUs.
- **Catalyst:** 2027-03-31: ReElement financing (judgement)
- **Key risks:** going-concern history, coal legacy liabilities, related-party complexity. Dilution risk: HIGH.
- **Kill condition:** Going-concern qualification in the next 10-K
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** sec.gov/cgi-bin/browse-edgar?a...mp;type=10-K

12. ASP ISOTOPES (ASPI)

- **Chain:** Nuclear medicine and advanced reactors → enriched isotopes / HALEU → ASPI aerodynamic separation (South Africa)
- **Verified fact:** Company claims (10-K, not re-read this pass): isotope enrichment plants in Pretoria; HALEU via QLE.
- **Catalyst:** 2027-06-30: Commercial isotope deliveries (judgement)
- **Key risks:** technology claims, short-seller reports in 2025, dilution. Dilution risk: HIGH.
- **Kill condition:** No audited isotope revenue growth in FY2026 10-K
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** sec.gov/cgi-bin/browse-edgar?a...mp;type=10-K

13. TUNGSTEN WEST (TUN.L)

- **Chain:** EU/UK tungsten processors → WO3 concentrate → Hemerdon (Devon), one of the few large Western tungsten deposits
- **Verified fact:** UK National Wealth Fund £71m package (£36m equity, £25m loan, £10m accordion); processing started Sep 2026; full production targeted Q1 2027.
- **Catalyst:** 2027-03-31: Hemerdon full commissioning target (Q1 2027)
- **Key risks:** Hemerdon failed once before (Wolf Minerals, 2018) on recoveries and noise permits; NWF holds heavy equity → dilution overhang. Dilution risk: HIGH.
- **Kill condition:** Commissioning slips past Q2 2027 or recoveries below plan in two consecutive updates
- **Contrarian:** Partly crowded. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** nationalwealthfund.org.uk/news...sten-supply/ · investormeetcompany.com/compan...4876733/view

14. TRI CHEMICAL LABORATORIES (4369.T)

- **Chain:** SK hynix (~80% of Hf precursor via SK Tri Chem JV) and Samsung (~20% since 2025) → high-k precursor → Tri Chemical
- **Verified fact:** High-k revenue ¥9.42bn (FY1/25); SK hynix diversifying Hf precursor suppliers (The Elec, trade press, not primary).

- **Catalyst:** 2026-12-12: Q3 FY1/27 results (judgement: mid-Dec)
- **Key risks:** SK hynix dual-sourcing erodes share; customer split not disclosed by the company. Dilution risk: MEDIUM.
- **Kill condition:** SK hynix share for SK Tri Chem reported below 60%
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** thelec.net/news/articleView.html?idxno=12207 · trichemical.com/english/ir/pdf/2025031402.pdf

15. IONIC RARE EARTHS (IXR.AX)

- **Chain:** EU/UK magnet users → recycled NdPr/Dy/Tb oxides → Belfast plant; Makuutu feed
- **Verified fact:** Company claims (not re-verified this pass): Belfast magnet recycling demonstration; Makuutu mining licence.
- **Catalyst:** 2027-03-31: Belfast scale-up (judgement)
- **Key risks:** tiny scale; serial dilution; Uganda jurisdiction. Dilution risk: HIGH.
- **Kill condition:** No commercial-scale funding by end-2027
- **Contrarian:** Not crowded on price. **Quant Validation:** VETO for recommendation-grade (no backtested KEPT signal; WATCH only).
- **Evidence:** ionicre.com/

04 Kills and exclusions

- **Already re-rated past the size filter (excluded, but they show the pattern):** AXT (InP substrates, $3.4bn), USA Rare Earth ($3.6bn), MP Materials ($6.8bn), Centrus ($2.1bn), Perpetua ($2.2bn), 5N Plus (C$2.4bn), ACM Research. The bottleneck trade in these names is mostly done; they are context, not candidates.
- **Killed on evidence:** Pioneer Power (no chokepoint), KULR (no chokepoint), Standard Lithium and Sigma (lithium is in surplus), Westwater (no customer, serial dilution), Altech/Aruma/Krakatoa (sub-cent, no asset under a dated catalyst).
- **Chokepoints with no small listed pure play found:** HBM itself, CoWoS/ABF substrates (Ajinomoto, Ibiden and Unimicron are large caps), glass substrates (private), photomasks (Photronics is mid-cap), high-purity quartz (Sibelco and The Quartz Corp are private), neon, GOES electrical steel, gas turbines, shipyards, solid rocket motors and energetics, BIOSECURE-driven CDMO.

05 Gaps

- No 13F/large-holder crowding pull this pass (Crustdata and Apify not used, $0 spent); crowding is price-based only. Analyst coverage counts not pulled. Google Trends and X mention checks not run.
- Japan EDINET API needs a key; JEM's customer data came from the published yuho PDF.
- Grade C names (Tri Chemical, Ionic, ASP Isotopes, American Resources) need primary re-verification before they move up.
- UN Comtrade monthly China exports (antimony, Ga/Ge, graphite, rare earths, tungsten) are being ingested; not yet joined to the catalysts.
- Ledger: 15 WATCH calls appended (hash chain verified = (True, 56)). They are scored forward from 23 Oct 2026 and anchored daily.

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